RAJALAKSHMI ENGINEERING COLLEGE
RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23334 Fundamentals of Data Science
Laboratory Record Note Book

Name :	ARAVINDH. P
Year / Branch / Section :	II/BE 'CSE' / 'J'
University Register No. :	211625070106 7
College Roll No. :	25070106 7
Semester :	III
Academic Year :	2026 - 2027

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS23334 Fundamentals of Data Science

NAME	**ARAVINDH I**
ROLL NO	**2116250701067**
YEAR	**II**
SEC	**J**

RAJALAKSHMI ENGINEERING COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: ..... Aravindh ... T .....

Academic Year: 2026-2027 **Semester:** ....III..... **Branch:** BE ... CSE ..

Register No. 2116 250701067

Certified that this is the bonafide record of work done by the above student in the Fundamentals of Data Science ***Laboratory during the academic year 2026 – 2027.***

Signature of Faculty in-charge

Submitted for the Practical Examination held on 05/12/2026.....

Internal Examiner

External Examiner

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S.NO	Experiment	Date	GitHub
1.a	Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib /seaborn for visualization.	24/07/2026	
1.b	Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.	24/07/2026	
1.c	Conduct an experiment to differentiate Structured, Un-structured and Sem istructured data based on dataset given.	24/07/2026	
1.d	Conduct an experiment to encrypt and decrypt given sensitive data Use the cryptography library fernet.	24/07/2026	
2	Upload and Analyse the data set given in csv format and perform data preprocessing and visualization.	24/07/2026	
3.a	Conduct an experiment to show data visualization using line plot.	31/07/2026	
3.b	Conduct an experiment to show data visualization using bar chart.	31/07/2026	
3.c	Conduct an experiment to show data visualization using pie chart.	31/07/2026	
4	To Count the frequency of occurrence of a word in a body of text is often needed during text processing.	07/08/2026	
5	Data Collection and Initial Exploration.	07/08/2026	
6	Time Series Analysis of Website Traffic.	14/08/2026	
7	Random Sampling and Sampling Distribution.	14/08/2026	
8	Conduct Z test for the Data given using Python.	21/08/2026	
9	Experiment to understand Matplotlib library use cases in Data Science through visualization.	21/08/2026	
10	Experiment to understand array function in Data science.	28/08/2026	
11	Experiment to understand pandas library use cases in Data science.	28/08/2026	
12	Experiment to detect outliers in a given data set.	11/09/2026	
13	Experiment to handle missing values and inappropriate data in a given data set.	18/09/2026	
14	Experiment to understand feature scaling.	18/09/2026	
15	Experiment to understand the data preprocessing in Data science.	25/09/2026	
16	Experiment to understand EDA-Quantitative and Qualitative analysis.	05/10/2026	
17	Experiment to understand Linear Regression for a given data set.	05/10/2026	
18	Experiment to understand KNN algorithm for given data set.	09/10/2026	
19	Experiment to understand Logistic Regression for a given data set.	09/10/2026	
20	Experiment to understand K-means clustering algorithm for a given data set.	09/10/2026	